

Explainer: Bonus Problem 2 (Rudin 7.21) — Isometric Embedding into $C(X)$

What's going on?

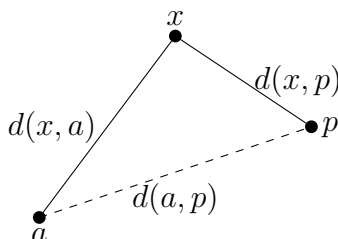
Given *any* metric space (X, d) , we embed it isometrically into the Banach space $C(X)$ of bounded continuous functions on X (with the sup norm). Then taking the closure gives a **completion** of X .

This is a very general and elegant construction: every metric space lives inside a complete one.

The embedding

Fix a base point $a \in X$. For each $p \in X$, define $f_p : X \rightarrow \mathbb{R}$ by:

$$f_p(x) = d(x, p) - d(x, a).$$



$f_p(x)$ measures “how much closer is x to a than to p ?” Subtracting $d(x, a)$ centers things so that $f_p(a) = 0 - 0 = 0$ always... wait, $f_p(a) = d(a, p) - d(a, a) = d(a, p)$. The subtraction of $d(x, a)$ ensures f_p is bounded.

Step 1: $|f_p(x)| \leq d(a, p)$

By the triangle inequality:

$$d(x, p) - d(x, a) \leq d(a, p) \quad \text{and} \quad d(x, a) - d(x, p) \leq d(a, p).$$

Both come from $d(x, p) \leq d(x, a) + d(a, p)$ (and the symmetric version). So $|f_p(x)| \leq d(a, p)$ for all x .

Since d is continuous (it's a metric!), f_p is continuous and bounded $\Rightarrow f_p \in C(X)$.

Step 2: $\|f_p - f_q\| = d(p, q)$

Compute:

$$(f_p - f_q)(x) = [d(x, p) - d(x, a)] - [d(x, q) - d(x, a)] = d(x, p) - d(x, q).$$

The $d(x, a)$ terms cancel! So:

$$\|f_p - f_q\| = \sup_{x \in X} |d(x, p) - d(x, q)|.$$

Upper bound: Triangle inequality gives $|d(x, p) - d(x, q)| \leq d(p, q)$.

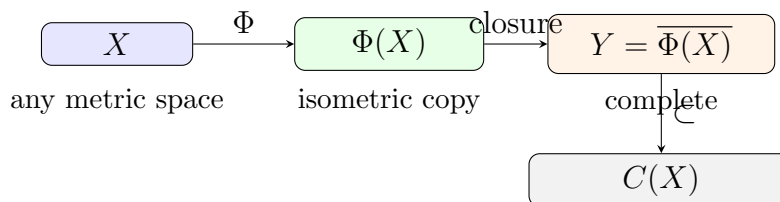
Lower bound: Set $x = p$: $|d(p, p) - d(p, q)| = d(p, q)$. Or $x = q$: same thing.

So $\|f_p - f_q\| = d(p, q)$. The map $\Phi : p \mapsto f_p$ is an **isometry**.

Step 3: The closure $Y = \overline{\Phi(X)}$ is complete

$C(X)$ with the sup norm is a complete metric space (Banach space). A closed subset of a complete space is complete. Y is closed by definition (it's a closure). Hence Y is complete.

Summary of the construction



X is isometric to $\Phi(X)$, which is dense in Y , and Y is complete. So Y is a completion of X .

Proof outline

1. Triangle inequality $\Rightarrow |f_p(x)| \leq d(a, p)$, so $f_p \in C(X)$.
2. Compute $f_p - f_q$, triangle inequality gives $\leq d(p, q)$, evaluation at $x = q$ gives $= d(p, q)$.
3. $C(X)$ is complete (standard), closed subsets of complete spaces are complete.